



10 – Set theory

Week 11 – 28 Oct 2025

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Cheat sheet

Set Identities

Theorem 6.2.2 Set identities

Let all sets referred to below be subsets of a universal set U .

1. *Commutative Laws:* For all sets A and B ,

$$(a) A \cup B = B \cup A \quad \text{and} \quad (b) A \cap B = B \cap A.$$

2. *Associative Laws:* For all sets A, B and C ,

$$(a) (A \cup B) \cup C = A \cup (B \cup C) \quad \text{and} \\ (b) (A \cap B) \cap C = A \cap (B \cap C).$$

3. *Distributive Laws:* For all sets A, B and C ,

$$(a) A \cup (B \cap C) = (A \cup B) \cap (A \cup C) \quad \text{and} \\ (b) A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

Set Identities

4. *Identity Laws:* For all sets A ,

$$(a) A \cup \emptyset = A \quad \text{and} \quad (b) A \cap U = A.$$

5. *Complement Laws:* For all sets A ,

$$(a) A \cup A^c = U \quad \text{and} \quad (b) A \cap A^c = \emptyset.$$

6. *Double complement Law:* For all sets A ,

$$(A^c)^c = A.$$

7. *Idempotent Laws:* For all sets A ,

$$(a) A \cup A = A \quad \text{and} \quad (b) A \cap A = A.$$

Set Identities

8. *Universal Bound Laws:* For all sets A ,

$$(a) A \cup U = U \quad \text{and} \quad (b) A \cap \emptyset = \emptyset.$$

9. *De Morgan's Law:* For all sets A and B ,

$$(a) (A \cup B)^c = A^c \cap B^c \quad \text{and} \quad (b) (A \cap B)^c = A^c \cup B^c.$$

10. *Absorption Laws:* For all sets A and B ,

$$(a) A \cup (A \cap B) = A \quad \text{and} \quad (b) A \cap (A \cup B) = A.$$

11. *Complements of U and $\emptyset$*

$$(a) U^c = \emptyset \quad \text{and} \quad (b) \emptyset^c = U.$$

Cheat sheet

Commutative laws	$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$
Associative laws	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
Distributive laws	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
Identity laws	$p \wedge 1 \equiv p$ $q \vee 0 \equiv q$
Negation laws	$p \vee \neg p \equiv 1$ $p \wedge \neg p \equiv 0$
Double negation	$\neg(\neg p) \equiv p$
Idempotent laws	$p \wedge p \equiv p$ $p \vee p \equiv p$
Universal bound laws	$p \vee 1 \equiv 1$ $p \wedge 0 \equiv 0$
De Morgan's law	$\neg(p \vee q) \equiv \neg p \wedge \neg q$
Absorption laws	$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$

Set Identities	
Identity laws	$A \cup \emptyset = A$ $A \cap U = A$
Complement laws	$A \cup A^c = U$ $A \cap A^c = \emptyset$
Double complement law	$(A^c)^c = A$
Idempotent laws	$A \cup A = A$ $A \cap A = A$
Universal Bound law	$A \cup U = U$ $A \cap \emptyset = \emptyset$
complements of U & $\emptyset$	$U^c = \emptyset$ $\emptyset^c = U$
set difference law	$A - B = A \cap B^c$

Cheat sheet

Definition

Given a set A , the **power set** of A , denoted $\mathcal{P}(A)$ is the set of all subsets of A .

$$A = \{a, b, c\}$$

$$\begin{aligned} P(A) = \{ & \{\}, && \leftarrow 0 \text{ elements} \\ & \{a\}, \{b\}, \{c\}, && \leftarrow 1 \text{ element} \\ & \{a,b\}, \{b,c\}, \{c,a\}, && \leftarrow 2 \text{ elements} \\ & \{a,b,c\} && \leftarrow 3 \text{ elements} \\ & \} \end{aligned}$$

Then you double check if they have listed EVERY subsets.

We know that $|P(A)| = 2^n$, so we use this to check.

Ex. 5.2.12 Prove the following by mathematical induction:

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{n(n+1)} = \frac{n}{n+1}$$

for all integers $n \geq 1$.

Ex. 5.3.17 Prove the following by mathematical induction:

$$1 + 3n < 4^n \text{ for every integer } n > 1.$$

Ex. 6.1.4. Let $A = \{n \in \mathbb{Z} \mid n = 5r \text{ for some integer } r\}$ and $B = \{m \in \mathbb{Z} \mid m = 20s \text{ for some integer } s\}$.

- a. Is $A \subseteq B$? Explain.
- b. Is $B \subseteq A$? Explain.

SOLUTION . a. No. Consider $5 \in \mathbb{Z}$. Since $5 = 5 \cdot 1$, and 1 is an integer, we know that $5 \in A$.

Suppose $5 \in B$. Then $5 = 20s$ for some integer s . Dividing both sides by 20 gives $s = \frac{5}{20} = \frac{1}{4}$ which is not an integer. This contradicts s is an integer, and so $5 \notin B$.

- b. Yes. Let m be an integer in B . Then $m = 20s$ for some integer s . We may express $m = 5(4s)$ and clearly, $4s$ is also an integer. Therefore $m \in A$.



Definition

A collection of nonempty sets $\{A_1, A_2, \dots, A_n\}$ is a **partition** of a set A if and only if

1. $A = A_1 \cup A_2 \cup \dots \cup A_n$,
2. $A_1, A_2, \dots, A_n$ are mutually disjoint.

Ex. 6.1.29. Let $\mathbb{R}$ be the set of all real numbers. Is $\{\mathbb{R}^+, \mathbb{R}^-, \{0\}\}$ a **partition** of $\mathbb{R}$? Explain your answer.

SOLUTION . Yes. This is because every real number x satisfies exactly one of the conditions:
 $x > 0$ or $x = 0$ or $x < 0$. □

AND they are mutually disjoint.

Ex. 6.1.33.

- a. Find $\mathcal{P}(\emptyset)$.
- b. Find $\mathcal{P}(\mathcal{P}(\emptyset))$.
- c. Find $\mathcal{P}(\mathcal{P}(\mathcal{P}(\emptyset)))$.

SOLUTION . a. $\mathcal{P}(\emptyset) = \{\emptyset\}$.

b. $\mathcal{P}(\mathcal{P}(\emptyset)) = \{\emptyset, \{\emptyset\}\}$.

c. $\mathcal{P}(\mathcal{P}(\mathcal{P}(\emptyset))) = \{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}, \{\emptyset, \{\emptyset\}\}\}$.

Note:

If it's too confusing, we can replace the various sets with other variables

Eg.

$$A = \{\{\}\}$$

Ex. 6.2.19. Use an element argument to prove the following statements. Assume that all sets are subsets of a universal set U .

For all sets A , B , and C ,

$$A \times (B \cap C) = (A \times B) \cap (A \times C).$$

Ex. 6.2.31, 35. Use the element method for proving a set equals the empty set to prove each of the following statements. Assume that all sets are subsets of a universal set U .

31. For all sets A , B , and C ,

if $A \subseteq B$ and $B \cap C = \emptyset$, then $A \cap C = \emptyset$.

35. For all sets A , B , and C ,

if $B \cap C \subseteq A$, then $(C - A) \cap (B - A) = \emptyset$.

Every time we get sets proofs questions, we use **element method**.

Ex. 6.3.20. Prove the statement if it is true and find a counterexample if it is false. Assume all sets are subsets of a universal set U .

For all sets A and B , $\mathcal{P}(A \cap B) = \mathcal{P}(A) \cap \mathcal{P}(B)$.

Ex. 6.3.34. Construct an algebraic proof for the given statement. Cite a property from Theorem 6.2.2 for each step.

For all sets A and B , $A - (A - B) = A \cap B$.

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Set Identities	
Identity laws	$A \cup \emptyset = A$ $A \cap U = A$
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Universal Bound law	$A \cup U = U$ $A \cap \emptyset = \emptyset$
complements of U & $\emptyset$	$U^c = \emptyset$ $\emptyset^c = U$
set difference law	$A - B = A \cap B^c$

Ex. 6.3.42. Simplify the given expression. Cite a property from Theorem 6.2.2 for each step.

$$(A - (A \cap B)) \cap (B - (A \cap B)).$$

SOLUTION . For some universal set U containing A and B ,

$(A - (A \cap B)) \cap (B - (A \cap B))$	
$= (A \cap (A \cap B)^c) \cap (B \cap (A \cap B)^c)$	by the set difference law
$= (A \cap (A^c \cup B^c)) \cap (B \cap (A^c \cup B^c))$	by De Morgan's law
$= ((A \cap A^c) \cup (A \cap B^c)) \cap ((B \cap A^c) \cup (B \cap B^c))$	by the distributive law
$= (\emptyset \cup (A \cap B^c)) \cap ((B \cap A^c) \cup \emptyset)$	by the complement law for $\cap$
$= ((A \cap B^c) \cup \emptyset) \cap ((B \cap A^c) \cup \emptyset)$	by the commutative law
$= (A \cap B^c) \cap (B \cap A^c)$	by the identity law for $\cup$
$= A \cap (B^c \cap (B \cap A^c))$	by the associative law for $\cap$
$= A \cap ((B^c \cap B) \cap A^c)$	by the associative law for $\cap$
$= A \cap (\emptyset \cap A^c)$	by the complement law for $\cap$
$= A \cap (A^c \cap \emptyset)$	by the commutative law for $\cap$
$= A \cap \emptyset$	by the universal bound law for $\cap$
$= \emptyset$	by the universal bound law for $\cap$.